isolation_forest Â¶

IsolationForestConfig Â¶

```
IsolationForestConfig(ntrees: int = 50, sample_size: int = 100, min_rows: int = 1, max_depth: int = 8, sample_rate: float = -1.0, mtries: int = -1, nbins: int = 2, seed: int | None = None, ignored_fields: set[str] | None = None, schema: Schema | None = None)
```

Bases: `TrainingConfig`

The H2O Isolation Forest configuration object used for model training.

This is an unsupervised learning algorithm used for anomaly detection. The target variable will be automatically excluded from features during training.

Parameters:

Name	Type	Description	Default
`ntrees` Â¶	`int`	Number of trees in the forest, defaults to 50.	`50`
`sample_size` Â¶	`int`	Number of samples to use per tree, defaults to 100.	`100`
`min_rows` Â¶	`int`	Minimum number of rows in a leaf, defaults to 1.	`1`
`max_depth` Â¶	`int`	Maximum depth of trees, defaults to 8.	`8`
`sample_rate` Â¶	`float`	Row sampling rate. Use -1 for automatic, defaults to -1.0.	`-1.0`
`mtries` Â¶	`int`	Number of features to try per split. Use -1 for automatic, defaults to -1.	`-1`
`nbins` Â¶	`int`	Number of bins for numerical features, defaults to 2.	`2`
`seed` Â¶	`int`	The seed value.	`None`
`ignored_fields` Â¶	`set of str`	Set of names corresponding to the fields to be ignored when training a model.	`None`

Name	Type	Description	Default
`schema` Â¶	[Schema]	Pass a schema to enable conversion between descs and internal names. By not passing a schema, ds-api will assume field names are references to internal Pulse names, not the ones in the UI.	`None`

Examples:

```
>>> model_config = IsolationForestConfig(
...     ntrees=100,
...     sample_size=500,
...     max_depth=10
... )
>>> model_config.ntrees # gets the ntrees: 100
>>> model_config.ntrees = 50 # sets the ntrees to 50
>>> model = model_project.train(
...     "model_name",
...     datasource,
...     "target_field",  # Will be excluded from features automatically
...     model_config
... )
>>> model.waitUntilModelTrained()
>>> config_from_model = model.get_config()
```

ignored_fields `property` `writable` Â¶

```
ignored_fields
```

max_depth `property` `writable` Â¶

```
max_depth
```

min_rows `property` `writable` Â¶

```
min_rows
```

mtries `property` `writable` Â¶

```
mtries
```

nbins `property` `writable` Â¶

```
nbins
```

ntrees `property` `writable` Â¶

```
ntrees
```

oversampling `property` `writable` Â¶

```
oversampling
```

sample_rate `property` `writable` Â¶

```
sample_rate
```

sample_size `property` `writable` Â¶

```
sample_size
```

seed `property` `writable` Â¶

```
seed
```

subclasses `class-attribute` `instance-attribute` Â¶

```
subclasses: List[JavaWrapper] = []
```

undersampling `property` `writable` Â¶

```
undersampling
```

auto_cast `staticmethod` Â¶

```
auto_cast(config: JavaWrapper) -> IsolationForestConfig
```

Creates an IsolationForestConfig from the java object.

Parameters:

Name	Type	Description	Default
`config` Â¶	`JavaWrapper`	Java configuration object.	*required*

Returns:

Type	Description
`IsolationForestConfig`	Configuration object according to its python representation.